

Google Summer of Code 2025 Proposal

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Synopsis

This proposal aims to enhance the Robotics Academy platform by introducing ROS 2-native solutions for existing. Currently, Robotics Academy allows students to solve exercises using a Hardware Abstraction Layer (HAL) without directly interacting with ROS 2 topics and services. However, many students prefer learning with native ROS 2 interfaces, making this an opportunity to expand their educational experience.

I propose adapting the **"Follow Line"** exercise to be fully ROS 2-based, replacing the HAL abstraction with direct use of ROS 2 topics and services. Additionally, I propose introducing a **"Basic Vacuum Cleaner"** exercise that will teach students how to work with scan data, odometry, and basic navigation. While this may exceed the 90-hour scope (estimated 105 hrs), it adds significant educational value by covering broader real-world robotics skills.

Benefits to the Community

- **Bridging the Gap Between HAL and ROS 2:** Currently, students can solve exercises without knowing ROS 2. This project allows them to learn real ROS 2 communication patterns, preparing them for real-world robotic systems.
- **Enhancing Educational Value:** By exposing students to image processing, scan data, odometry, and navigation, the exercises will strengthen Robotics Academy's curriculum.
- **Aligning Exercises with Real-World ROS Practices:** Refactoring existing exercises to use native ROS 2 interfaces will make them more realistic, helping students transition smoothly from academic learning to practical robotics development.

Deliverables

- **Follow Line (ROS 2 Version)**

The goal of this exercise is to perform a PID reactive control capable of following a red line painted on a racing circuit. The line is detected using the camera image received as a `sensor_msgs/msg/Image` message from the `/camera/image_raw` topic. The goal of this task is to remove the HAL and make the exercise function entirely using native ROS 2 topics and services.

This exercise introduces:

- ROS 2 publishers and subscribers
- `cv_bridge` for converting ROS 2 images
- Publishing velocity using `geometry_msgs/msg/Twist`

The following improvements will be implemented:

- **Fix Execution Flow:** Remove the existing while loop and replace it with ROS 2 callback mechanisms using `rclpy.spin()`. The user will be responsible for defining `rclpy.spin()`, ensuring they learn proper ROS 2 practices while building applications. The goal is to make the experience as close as possible to working with actual ROS, so that users can seamlessly transition to real-world ROS applications.
- **Fix Image Display in GUI:** Currently, there is an issue with obtaining the image feed from the image publisher. This needs to be fixed to ensure that the camera feed is correctly displayed using `GUI.showImage()`.
- **Enable ROS 2 Debugging Tools:** Ensure compatibility with standard ROS 2 debugging tools such as `ros2 topic list`, `ros2 topic echo`, `ros2 node info` and visual tools like `rqt_graph`. Integrating these tools will allow users to debug their applications using methods commonly used in ROS, improving their understanding of ROS 2 troubleshooting.
- **Improve Visualization:** Investigate and implement support for `rviz2` and `rqt_image_view` to enhance debugging and provide better visualization of the robot's perception and environment.

Expected Outcome: By the end of this deliverable, a ROS 2-native version of the Follow Line exercise will be created, removing the need for HAL. Users will be able to implement logic using standard ROS 2 practices—defining nodes, publishers, and subscribers. ROS 2 debugging tools like `ros2 topic echo`, `ros2 node info`, and visual tools like `rqt_graph`, `rqt_image_view` will be supported. The user will be able to follow the line on the F1 circuit using code written entirely with native ROS 2.

- **Basic Vacuum Cleaner (ROS 2 Version)**

The objective of this practice is to implement the logic of a navigation algorithm for an autonomous vacuum cleaner. The main goal is to cover the largest area of the simulated house using the programmed logic.

This exercise will introduce students to:

- Processing LiDAR scan data from /scan
- Using odometry data to track position
- Implementing obstacle avoidance and simple coverage patterns
- Hands-on experience with ROS 2 topics and parameters

The following improvements will be implemented (if time permits):

- **Remove HAL Dependencies:** Transition the exercise to use only ROS 2 topics and services for LiDAR and movement control (cmd_vel).
- **Fix ROS 2 Execution Flow:** Replace the existing while loop with rclpy.spin() for event-driven execution, following the same approach as in Follow Line.
- **Fix & Enhance LiDAR Visualization:** Ensure /scan topic data is visualizable in rviz2; explore alternatives for LiDAR display in the Robotics Academy GUI if needed for user-side development.

Expected Outcome: By the end of this deliverable, a ROS 2-native version of the Basic Vacuum Cleaner exercise will be created, eliminating the need for HAL. Users will implement logic using standard ROS 2 practices—processing LiDAR and odometry data via native topics, and publishing velocity commands. Tools like ros2 topic echo, ros2 node info, and rviz2 will be supported for debugging and visualization. The robot will autonomously navigate and cover the simulated house area using logic written entirely with ROS 2.

Learning Outcomes for End User

By the end of both exercises, the user will:

- Learn to write complete logic using native ROS 2 concepts like nodes, topics, publishers, and subscribers.
- Understand how to work with different types of sensor data including camera (/camera/image_raw), LiDAR (/scan), and odometry (/odom).
- Learn how to publish Twist messages for robot movement using geometry_msgs/msg/Twist.
- Get familiar with ROS 2 debugging tools such as ros2 topic echo, ros2 node info, ros2 topic list.
- Use visualization tools like rviz2, rqt_graph, and rqt_image_view to observe and debug system behavior.
- Follow proper ROS 2 execution flow using rclpy.spin() and callback-based programming.
- Gain hands-on experience building real-world robotic behaviors like line following and obstacle avoidance using scan data.

Together, these exercises build a strong foundation for learners to confidently work with ROS 2 in practical robotics scenarios.

Timeline

Community Bonding Period (Pre-coding, 2 Weeks)

- Set up the development environment for ROS 2 exercises. Discuss and finalize the migration approach with mentors. Identify potential issues with the HAL-to-ROS 2 conversion.

Follow Line Migration (50 hrs approx.)

- **Phase 1 (June 2 – June 10, 2025) (Coding efforts: 15 hrs approx.):**
 - Convert *Follow Line* exercise to use ROS 2 topics instead of HAL.
 - Fix the **while loop issue** so the control flow is event-driven.
- **Phase 2 (June 11 – June 19, 2025) (Coding efforts: 15 hrs approx.):**
 - Ensure images are displayed correctly using `GUI.showImage()`.
 - Integrate basic ROS 2 debugging tools (`ros2 topic list`, `ros2 node info`).
- **Phase 3 (June 20 – June 24, 2025) (Coding efforts: 5 hrs approx.):**
 - Add support for `rqt_graph` or similar visualization tools if feasible.
 - Conduct initial tests and debugging.
- **Phase 4 Midterm Goal (June 25 – July 4th, 2025) (Coding efforts: 15 hrs approx.):**
 - Finalize *Follow Line* migration.
 - Write documentation on how students should use the updated exercise.
 - If time permits, explore displaying images in `rqt_image_view`.

Basic Vacuum Cleaner (If Time Permits) (55 hrs approx.)

- **Phase 1 (July 5 – July 12, 2025) (Coding efforts: 10 hrs approx.):**
 - Implement a ROS 2 version of *Basic Vacuum Cleaner*, similar to Follow Line.
 - Replace HAL with ROS 2 topics (`/scan`, `odometry`, etc.).
- **Phase 2 (July 13 – July 21, 2025) (Coding efforts: 15 hrs approx.):**
 - Fix visualization issues for the vacuum cleaner task, focusing on LiDAR data (using `rviz2` for real-time scan visualization).
- **Phase 3 (July 22 – July 27, 2025) (Coding efforts: 15 hrs approx.):**
 - Optimize and test *Basic Vacuum Cleaner*.
 - Conduct mentor/community feedback review.
- **Phase 4 (July 28 – August 5, 2025) (Final Phase) (Coding efforts: 15 hrs approx.):**
 - Improve documentation and usability.
 - Submit final PRs and write a project summary.

Contributions to JdeRobot

I have actively contributed to JdeRobot's Robotics Academy, submitting bug reports, code fixes, and improvements. Below are my key contributions:

Contribution Type	Description	Link
Issue	Server does not fully terminate on quitting server	#2958
Merged PR	Added images for Montecarlo Laser Loc as mentioned in #2744	#2992
Merged PR	Fixed Marker Visual Loc Robot goes to another pose on reset / play as mentioned in #2911	#2995
Discussions	Service robot Pose Fix Enhancement	#3000
Issue	[Basic Computer Vision / Color Filter] Page not Found (gh-pages)	#3001
Merged PR	Fixed prototype to running and broken link under computer vision section for basic computer vision	#3002

These contributions have helped me gain a deep understanding of Robotics Academy's architecture and workflow, which will be beneficial in successfully implementing this GSoC project.

Experience & Open-Source Contributions

- 2+ years working with ROS and robotics frameworks
- Experience contributing to Robotics Academy (Issues raised, PRs submitted, and fixes implemented)
- Space Navigation Intern @ Indian Space Research Organization (ISRO) – (1st April to 15th May)
- Ex intern @ Centre of Artificial Intelligence and Robotics, DRDO
- Taken part and successfully completed Hacktober Fest 2020
- 5x Hackathon Wins, Top 10 in E-yantra Robotics Challenge 2025
- Tech Lead @ Google Developers Group MVJCE & Ex - Vice President of Robo Club MVJCE

Why Me?

- Passionate about robotics education and open-source contributions.
- Already familiar with Robotics Academy, its architecture, and existing exercises.
- Strong experience in ROS 2-based robotics development and simulation.
- Committed to long-term contributions beyond GSOC, with the goal of gradually taking on more responsibilities within the organization and eventually contributing as a core maintainer.

Why JdeRobot?

- **Strong Alignment with My Interests:** JdeRobot's commitment to robotics education and open-source learning resonates with me.
- **Active User and Advocate:** I not only use Robotics Academy for my own learning but have also recommended it to my juniors and helped onboard them to the platform.
- **Excited to Expand ROS 2 Capabilities:** I see a great opportunity to contribute by improving ROS 2 integration in Robotics Academy.
- **Long-Term Contribution Goal:** Having already worked with JdeRobot's repositories, I want to deepen my involvement and eventually take on more responsibilities within the community.

GSoC Participation Questions

1. **Do you fully understand the commitment of taking part in GSoC? Do you have any conflicts during the proposed timeline?**

Yes, I fully understand that GSoC is equivalent to a full-time paid summer internship. I have checked my academic calendar and do not have any major conflicts. I am prepared to dedicate the required time to successfully complete the project.

2. **Is this your first time applying for GSoC? Are you submitting another proposal to a different organization for GSoC 2025?**

This is my first time applying for GSoC, and I am only submitting a proposal to JdeRobot.

3. **What do you expect from the GSoC program if selected? What is your post-GSoC intention?**

Through GSoC, I aim to deepen my ROS 2 expertise, contribute meaningfully to open-source, and refine my ability to develop structured robotics exercises. After GSoC, I plan to continue contributing to JdeRobot, mentoring future contributors, help maintain the Robotics Academy and hopefully become a core maintainer.